INMO(34^{th}) – 2019

Time – 4 Hrs

Instructions:

- **Calculators (in any form) and protractors are not allowed.**
- **Ruler and compass allowed.**
- **Answer all the questions.**
- **All Questions carry equal marks. Maximum marks: 102.**
- **Answer to each question should start on a new page. Clearly indicate the question number.**

1. Let ABC be a triangle with $\angle BAC > 90^0$. Let D be a point on the segment BC and E be a point on the line AD such that AD is a tangent to the circumcircle of triangle ACD at A and BE is perpendicular to AD. Given that CA = CD and AE = CE, determine $\angle BCE$ in degrees.
2. Let $A_1B_1C_1D_1E_1$ be a pentagon. For $2 \le n \le 11$, let $A_nB_nC_nD_nE_n$ be a pentagon whose vertices are the mid points of the sides $A_{n-1}B_{n-1}C_{n-1}D_{n-1}E_{n-1}$. All the 5 vertices of each of the 11 pentagons are arbitrarily coloured red or blue. Prove that four points among 55 points have the same colour and from the vertices of a cyclic quadrilateral.
3. Prove that $\gcd(m,n) + \gcd(m+1,n+1) + \gcd(m+2,n+2) \le 2|m-n| + 1$.
4. Let n and M be positive integer such that $M > n^{n-1}$. Prove that there are n distinct primes $p_1, p_2, ..., p_n$ such that p_j divides $M + j$ for $1 \le j \le n$.
5. Let AB be a diameter of a circle Γ and let V be a point on Γ different from A and B. Let D be the foot of the perpendicular from C on to AB. Let K be a point of the segment CD such that AC is equal to the semiperimeter of the triangle ADK. Show that the excircle of triangle ADK opposite A is tangent to Γ.
6. Let f be a function defined from the set $\{(x,y) : x, y \in R,\ xy \ne 0\}$ to the set of all positive real numbers such that:

   (a) $f(xy,z) = f(x,z)f(y,z),\ for\ all\ x, y \ne 0;$

   (b) $f(x,1-x) = 1,\ for\ all\ x \ne 0,1;$

   Prove that:

   (a) $f(x,x) = f(x,z-1) = 1,\ for\ all\ x \ne 0;$

   (b) $f(x,y)f(y,x) = 1,\ for\ all\ x, y \ne 0;$

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